

Jasdeep Singh

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EDUCATION:

University of Colorado at Boulder, College of Engineering and Applied Sciences
B.S. in Computer Science, B.S. in Applied Mathematics
Statistics Minor and Quantitative Finance Certificate

Expected Graduation May 2026
Boulder, CO

- **GPA 3.72** | Deans list 2021-present
- CU Esteemed Scholars-Sewall Scholarship
- Sikh Student Association, South Asian Student Association, Bole Boulder Bhangra, CU Quants, CU Traders

Engineering Honors Program

- Selective honors program recognizing top engineering students and requiring completion of an Honors Senior Thesis.

WORK EXPERIENCE:

The Menon Lab, University of Utah

May 2025 to Present

Undergraduate Research Intern

Salt Lake City, UT

- Investigating deep learning approaches for computational imaging, focusing on transformer-based and pix2pix architectures for translating EDoF images into varifocal image sets.
- Designed and 3D printed a custom housing for a Raspberry Pi and dual-camera system to collect experimental imaging data.
- Collected and analyzed datasets consisting of single EDoF images and corresponding varifocal image sequences, with ongoing work on data expansion and model refinement.

Jain Research Lab, Purdue University

May 2024 to Aug 2024

Undergraduate Research Intern

Lafayette, IN

- Modified an online drone simulation where participants provided self-confidence assessments after each trial and received feedback from a large language model.
- Conducted an open-loop experiment to collect data for training a Markov Decision Process (MDP) model, with random automation assistance and chatbot feedback.
- Closed the loop by training the MDP model to determine an optimal policy for providing feedback and automation assistance based on participant performance and confidence.

Programming Languages and Verification Laboratory, University of Colorado Boulder

May 2023 to Present

Undergraduate Researcher

Boulder, CO

- Developed a robustness evaluator in Scala to assess drone flight skills using signal temporal logic and dynamic programming for trajectory segmentation.
- Analyzed flight path data with Python and MATLAB, enabling trend identification and specialized feedback using machine learning techniques.
- Co-authored and presented a paper at Hybrid Systems Control and Computation 2024, showcasing the segmentation method and its impact on user feedback.

PROJECTS:

Chladni Plates

October 2022 to December 2022

- Conducted research on advanced differential equations, including the Double Laplacian & Bessel's equations
- Analyzed partial differential equations (PDEs) both analytically & numerically using Mathematica
- Co-authored academic report on modeling sand patterns on vibrating metal plate with PDEs

MiniGit

Jun 2022 to Aug 2022

- Developed version control system that tracks and manages file changes
- Allows users to add/remove files, search, commit, and checkout within repository
- Implemented data structures (e.g. linked-lists, hash tables) in C++ for project management

Image Compression

Mar 2022 to Jun 2022

- Researched and applied advanced linear algebra and analysis concepts (e.g. SVD, Fourier Transform)
- Analyzed image compression techniques using SVD and Fourier Transform in MATLAB
- Co-authored academic report on results of different image compression techniques

SKILLS AND INTERESTS:

Programming Languages: Python, C++, Java, R, Node.JS, SQL, MATLAB, Scala, Mathematica, HTML, CSS, PHP

Interests: Autonomous Systems, Hybrid Systems, Machine Learning, Deep Learning, Data Science, Optimization, Cyber Physical Systems, Formal Methods and Verification, Statistical Analysis, Numerical Computation, Software Engineering